

# CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Fourth Semester of B.Tech. (CE/IT) Examination

May 2011

## IT206 Object Oriented Programming using JAVA

Date: 13/05/2011

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 70

### Instructions:

1. Figure to the right indicates full marks
2. Make suitable assumption whenever necessary and mention it clearly
3. Parts of same question should be answered together and in the same sequence.
4. Answers to the sections must be written separately

### SECTION – I

Q:1

- |                                                                                                |   |
|------------------------------------------------------------------------------------------------|---|
| (a) Compare and contrast structured (procedural) programming with object oriented programming. | 3 |
| (b) What is the purpose of garbage collection in Java, and when is it used?                    | 2 |
| (c) What are the primary methods executed during the life cycle of an applet?                  | 2 |
| (d) Differentiate equals() and == operator for string class.                                   | 2 |
| (e) What is Method Overloading?                                                                | 2 |

Q:2

- |                                                                                 |   |
|---------------------------------------------------------------------------------|---|
| (a) Compare and contrast between interaction diagram and collaboration diagram. | 4 |
| (b) What is the purpose of producing use-case?                                  | 2 |
| (c) What is <i>instanceof</i> operator? Explain with suitable example.          | 4 |
| (d) Which is true? (Choose all that apply)                                      | 2 |

- i.) "X extends Y" is correct if and only if X is a class and Y is an interface.
- ii.) "X extends Y" is correct if and only if X is an interface and Y is a class.
- iii.) "X extends Y" is correct if X and Y are either both classes or both interfaces.
- iv.) "X extends Y" is correct for all combinations of X and Y being classes and/or interfaces.

OR

Q:2

- |                                                                                  |   |
|----------------------------------------------------------------------------------|---|
| (a) Compare and contrast between activity diagram and state chart diagram.       | 4 |
| (b) What does it mean to say that an object – oriented system is highly modular? | 2 |
| (c) What is Method Overriding? Explain with suitable example.                    | 4 |
| (d) What is the difference between static and non-static variables?              | 2 |

Q:3

- |                                                                                                                            |   |
|----------------------------------------------------------------------------------------------------------------------------|---|
| (a) What is a thread? Explain the life cycle of thread.                                                                    | 6 |
| (b) Explain following StringBuffer class method.<br>i.) insert()    ii.) indexOf()    iii.) setCharAt()    iv.) capacity() | 4 |
| (c) What are pass by value and pass by reference?                                                                          | 2 |

OR

Q:3

- |                                                                                                                             |   |
|-----------------------------------------------------------------------------------------------------------------------------|---|
| (a) What is an interface? Explain interface inheritance with suitable example.                                              | 6 |
| (b) Explain the following concepts<br>(i) forking and merging (ii) Branching and joining with respect to Activity diagram.. | 4 |
| (c) How attributes and operations of a class can be organized?                                                              | 2 |



## SECTION – II

- Q:4**
- |                                                                                                                                                                                       |   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| (a) What is interface? What are the default modifier for and interface, interface method and variable?                                                                                | 3 |
| (b) What is Explicit Type casting?                                                                                                                                                    | 2 |
| (c) If object obj1 can access object obj2 that is eligible for garbage collection, then obj1 is also eligible for garbage collection. Statement is true or false. Justify your answer | 2 |
| (d) Outline the main steps in developing a class diagram from a use-case diagram                                                                                                      | 4 |

- Q:5**
- |                                                                                                                                                                                                                          |   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| (a) Prepare a use-case model fro “ATM”.                                                                                                                                                                                  | 3 |
| (b) Write an applet that contains a checkbox group with three check boxes labeled “Red”, “Green”, and “Blue”. There is also a 40 X 60 pixel canvas. The selections of the check boxes determine the color of the canvas. | 4 |
| (c) What is the difference between final, finally and finalize?                                                                                                                                                          | 3 |
| (d) Explain the reason for each keyword of public static void main(String args[])?                                                                                                                                       | 2 |

OR

- Q:5**
- |                                                                                                                                                                                                      |   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| (a) Explain different way of using thread?                                                                                                                                                           | 3 |
| (b) Write an application that copies on character file to a second character file. The source file is the first command line argument, and the destination file is the second command line argument. | 4 |
| (c) What is the difference between diagram and model? What rules describe the relationship between a subclass and its superclass?                                                                    | 3 |
| (d) What is the difference between an Interface and an Abstract class?                                                                                                                               | 2 |

- Q:6 Answer the following questions (Any Three)**
- |                                                                               |    |
|-------------------------------------------------------------------------------|----|
| (a) Draw an activity diagram for “order processing” or “processing an order”. | 12 |
| (b) Write a short note on delegation event model.                             |    |
| (c) Write a short note on Inter-thread communication.                         |    |
| (d) Name three types of layout managers and briefly explain their operation.  |    |

OR

OR